



Systematics and evolutionary history of the genus *Micromys* (Mammalia: Rodentia: Muridae)

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Abstract

The genus *Micromys* is one of the smallest-sized rodents widely distributed in temperate regions of Eurasia. However, its taxonomy and evolutionary history remain unresolved. In this study, we analyzed the phylogenetic and morphological relationships of *Micromys* species using two mitochondrial genes, three nuclear genes, and a morphological dataset. We also analyzed its geographical distribution structure using 140 cytochrome b (*Cyt b*) gene sequences. Our results supported the existence of three species in the genus *Micromys*: *M. minutus*, *M. erythrotis*, and *M. pygmaeus*, which was previously considered a subspecies of *M. minutus* or *M. erythrotis*. *Micromys minutus* is widely distributed in Europe and North-East Asia, *M. erythrotis* is distributed in central and southern China, Vietnam, Myanmar, and India, and *M. pygmaeus* is restricted to the middle-to-high elevations of the Himalayas and Hengduan Mountains (1600–3800 m). Phylogenetic analyses indicated that *M. pygmaeus* is a sister species to *M. erythrotis*, with a *p*-distance of 11.4% (*Cyt b*) and 9.2% (*COI*) between them. The divergence between *M. minutus* and *M. pygmaeus* + *M. erythrotis* occurred ca. 3.35 Ma, while *M. pygmaeus* and *M. erythrotis* diverged ca. 2.55 Ma. The divergence of the *Micromys* species was likely affected by the rapid uplift of the Tibetan Plateau and Hengduan Mountains, and the global cooling and desiccating events at the Pliocene/Pleistocene boundaries.

Keywords *Micromys* · Taxonomy · Phylogeny · Small mammals · Himalayas · Hengduan mountains

Introduction

The Harvest mouse, genus *Micromys* Dehne, 1841, is one of the smallest-sized rodent taxa in the family Muridae. The animals are famous for their ability to build nests on the stems of wheat or other plants. It is widely distributed in temperate regions of Asia and Europe, ranging from the British Isles and the northern Iberian Peninsula through Europe, Russia, China, Korea, Japan, Myanmar, Laos, Vietnam, and NE India (Abramov et al. 2009; Wilson et al. 2017). *Micromys* species inhabit grasslands, farmlands, and forests, with a preference for wetter microhabitats such as forest edges and weedy shrubs along stream edges (Harris 1979).

The broad distributional range of *Micromys* makes it challenging to collect specimens, resulting in few studies and a lack of systematic resolution. To date, the taxonomic and distribution limits of *Micromys* are still controversial. The first recognized species of the genus was *Mus minutus*, described by Pallas (1771) based on specimens collected from Ulyanovsk, Russia (= Simbirsk). Since then, more than forty species/subspecies have been described under the

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genus, with most of them treated as synonyms of *M. minutus* (Wilson and Reeder 2005). Some authors have pointed out that *M. minutus* specimens from mainland Indomalaya (South China, North-east India, and North Vietnam) differed from those of Europe in pelage color and body proportions (Corbet and Hill 1992). A recent molecular study based on the mitochondrial cytochrome b (*Cyt b*) gene and control region revealed very low variation among populations, except for an extremely high level of genetic differentiation between a specimen from Chengdu, Sichuan, China, and all other populations (Yasuda et al. 2005). Abramov et al. (2009) found that the populations in northern Vietnam clustered with the Chengdu specimen, which was morphologically distinct from *M. minutus*; they considered this form as representing a distinct species, *M. erythrotis* Blyth 1856. However, Wilson et al. (2017) did not follow this assignment owing to the small sample size of Abramov et al. (2009). More recently, our preliminary study revealed that the *Micromys* populations from Anhui, eastern China, and Vietnam belonged to the same lineage, sympatric to *M. minutus* from Qingliang Mountain, Anhui, China (Pei et al. 2021), supporting *M. erythrotis* as a valid species. Notably, we further found that the two sequences from Pagès et al. (2010) deposited in GenBank (HM217360 and HM217361), which were sampled from China, revealed a high genetic distance with *M. minutus* and *M. erythrotis* (genetic distance of *Cyt b* > 11%), implying species diversity in *Micromys* is likely currently underestimated.

In recent years, we conducted extensive field surveys in western and southern China. We collected several *Micromys* specimens, including more than 30 specimens from the Himalayas and Hengduan Mountains that clustered with the sequences from Pagès et al. (2010), which are likely to represent an undescribed species (hereafter, *Micromys* sp.). These specimens allowed us to re-examine the taxonomy and evolutionary history of the genus *Micromys*. We sequenced two mitochondrial and three nuclear genes from these specimens and evaluated their morphological and morphometric variation. By integrating the GenBank data, we aimed to (1) clarify the taxonomy and distribution of *Micromys*; (2) analyze the phylogeny and diversification of taxa in the genus *Micromys*.

Material and methods

Morphological analysis

Specimens of *Micromys* used for this study were deposited at the Kunming Natural History Museum of Zoology, Kunming Institute of Zoology (KIZ), Chinese Academy of Sciences, Kunming, China, and Anhui Normal University (AHNU), Wuhu, China. A total of 45 *Micromys* specimens

were collected. Voucher information and location data are listed in Table S1. Animals were handled based on the animal care and use guidelines of the American Society of Mammalogists (Sikes and the Animal Care and Use Committee of the American Society of Mammalogists 2016). The experiments were performed following the guidelines and regulations approved by the internal review board of AHNU (with no special approval number) and with the permission of local authorities.

Specimens were identified following Abramov et al. (2009). To avoid misidentification, we reconstructed a *Cyt b* gene tree, included the sequences from GenBank, and carefully examined the morphological characters of individuals in each clade. The 45 *Micromys* specimens were assigned to two recognized and putative species, including nine *M. erythrotis*, five *M. minutus*, and 31 *Micromys* sp. (Table S1). Five external measurements: head and body length (HB), tail length (TL), Ear length (EL), hind foot length (HL), and weight (W) were measured in the field or recorded from specimen labels and are reported to the nearest 1 mm or g. However, these measurements were not used for morphological analyses due to considerable inter-observer variation (Chen et al. 2017).

For morphometric analyses, 16 craniomandibular variables were measured using a digital caliper graduated to the nearest 0.01 mm, following Yang et al. (2005). These craniodental metrics included: the greatest length of the skull (GLS), basal length (BL), palatoincise length (PIL), postpalatal length (PPL), zygomatic breadth (ZB), cranial breadth (CB), interorbital breadth (IOB), maximum width across the upper second molars (M^2 - M^2), rostral breadth (RB), upper toothrow length (UTL), upper molar row length (UML), nasal length (NL), cranial height (CH), foramen magnum breadth (BMF), mandibular length (ML), lower toothrow length (LTR). All the craniodental measurements were taken by the first author.

We used principal component analysis (PCA) to evaluate the morphological variation among populations of *Micromys*. The PCA was implemented in SPSS 19.0 (SPSS Inc., USA), and all craniodental measurements were first $\log_{10}$ -transformed. The terminologies for morphological descriptions followed Elbroch (2006) and Abramov et al. (2009).

Phylogenetic analyses and molecular divergence times

The total genomic DNA of 32 *Micromys* specimens was extracted from muscle or liver using a DNA extraction kit (Qiagen DNeasy Blood and Tissue Kit, Germany). Two mitochondrial genes (complete cytochrome b, *Cyt b* and Cytochrome c oxidase subunit 1, *COI*) and segments of 3 nuclear genes (retinol Binding Protein 3, *RBP3*;

recombination-activating gene 1, *RAG1*; and growth hormone receptor gene, *GHR*) were amplified. The primers and PCR program conditions are provided in Supporting Information (Table S2). The PCR products were sequenced in both directions using the BigDye Terminator Cycle Sequencing kit v. 3.1 (ThermoFisher Scientific, USA) on an ABI 3730xl sequencer. The DNA sequences were assembled using Seqman (DNASTAR, Lasergene v7.1). Corresponding sequences of a *Micromys* specimen (HS1148) from Japan were downloaded from GenBank and included in our dataset (Table S1). The sequences were aligned using MUSCLE (Edgar 2004), then checked manually. Sequences of *Mus pahari*, *Phloeomys pallidus*, *Batomys granti*, *Rattus exulans*, *Niviventer cremoriventer*, and *Apodemus agrarius* were also downloaded from GenBank and included in the alignments as outgroup taxa.

Three concatenated datasets were used for the phylogenetic analyses: (1) mitochondrial genes (*Cyt b* and *COI*), (2) nuclear genes, and (3) the mitochondrial-nuclear genes. We used Maximum-likelihood (ML) and Bayesian inference (BI) methods to reconstruct phylogenetic relationships. The ML phylogenies were inferred using IQ-TREE (Nguyen et al. 2015), with 5000 ultrafast bootstraps to estimate branch support. The BI phylogenies were reconstructed using MrBayes 3.2.6 (Ronquist et al. 2012) with two parallel runs and 2,000,000 generations to estimate posterior probabilities. The best-fit partitioning scheme and evolutionary models were selected using PartitionFinder 2 (Lanfear et al. 2017) with the greedy algorithm under the Bayesian information criterion (BIC) (Table S3). Both ML and BI analyses were performed in PhyloSuite (Zhang et al. 2020). The ultrafast bootstrap value (UFBoot) ≥ 95 and posterior probabilities (PP) ≥ 0.95 were considered strong supports (Huelsenbeck and Rannala 2004; Minh et al. 2013).

We estimated divergence time using BEAST v2.6.7 (Bouckaert et al. 2014) using the concatenated mitochondrial and nuclear gene dataset, with data blocks defined based on genes and codon positions. The evolutionary models and the best partition schemes were estimated using PartitionFinder 2 (Lanfear et al. 2017) under BIC (Table S3). Divergence dates were calibrated based on two secondary calibrations following Aghová et al. (2018): (1) the split of the most recent common ancestor of “core Murinae” set at 12.5 Ma (95% CI: 11.31–13.99 Ma), with a lognormal distribution prior (mean: 12.5, standard deviation: 0.07, offset: 0), such that the median age was at 12.5 Ma and the 95% CI was 11.1–14.0 Ma, and (2) the split of the most recent common ancestor Rattini set at 11.2 Ma (95% HPD: 9.91–12.67 Ma), also with a lognormal distribution prior (mean: 11.2, standard deviation: 0.08, offset: 0), such that the median age was 11.2 Ma and the 95% CI was 9.8–12.7 Ma. The analyses were implemented with 100 million generations, with chains sampled every 10,000 generations. Each analysis used a

lognormal relaxed clock model and a birth–death tree prior. The convergence of parameters was confirmed using Tracer v1.7 (Rambaut et al. 2018).

Species delimitation

To test the taxonomic status of *Micromys*, we first calculated the uncorrected *p*-distance of *Cyt b* and *COI* sequences in MEGA-X (Kumar et al. 2018) to determine the genetic variation between groups. Then we used two methods to infer molecular species delimitations. We first used Bayesian Phylogenetics and Phylogeography (BPP) method for species delimitation, with *M. minutus*, *M. erythrotis*, and *Micromys* sp. assigned as hypothetical species based on gene trees. The BPP analysis was performed using all five genes using BPP v3.1 (Yang 2015). To avoid false positives (over-splitting errors) due to an incorrect guide tree, we performed analyses with both A10 (Yang and Rannala 2010) and A11 (joint species delimitation and species tree estimation) (Yang and Rannala 2014) models. Moreover, we estimated the generalized mixed Yule-coalescent (GMYC) model based on the phylogenetic species concept, implemented with the R package *splits* (Pons et al. 2006). An ultrametric gene tree based on concatenated mitochondrial and nuclear datasets was used for the GMYC analysis, and outgroups were removed using the R package *ape* before analysis (Paradis and Schliep 2019).

Geographical distribution structure

To understand the population dynamics of *Micromys*, we first performed a phylogenetic analysis based on 140 *Cyt b* sequences (107 published sequences downloaded from NCBI GenBank and 33 new sequences from the current study; Table S4). We also inferred the phylogeographic structure of *Micromys* in DnaSP v5.10 (Librado and Rozas 2009) and PopART v1.7 software (Leigh and Bryant 2015) using the 140 *Cyt b* sequences. DnaSP was used to extract haplotypes and PopART to construct a median-joining haplotype network. We assembled a fossil record dataset of *Micromys* with data obtained from the Paleobiology Database (PBDB, <https://paleobiodb.org>), the New and Old Worlds Database of fossil mammals (NOW, <https://nowdatabase.luomus.fi>), the National Infrastructure of mineral rock and fossil resources for science and technology of China (<http://www.nimrf.net.cn/ept>) and Mindat (<https://www.mindat.org>) (Table S5).

Demographic history of *Micromys*

We used *Cyt b* sequences datasets to analyze the demographic history of the three putative species within *Micromys*. We calculated Tajima's *D* and Fu's *FS* to test for

neutrality and used the mismatch distribution to identify historical processes between recent demographic expansion and population equilibrium. These analyses were performed in Arlequin 3.5 (Excoffier and Lischer 2010).

Since the mismatch distribution and neutrality test may not fully extract historical signals, we further performed Bayesian skyline analysis to infer the historical demographic dynamics of each of the three putative species. We conducted the analyses in BEAST v2.6.7 using lognormal relaxed clocks (Bouckaert et al. 2014) and 10 million generations, with chains sampled every 1000 generations. Each analysis used a random starting tree and the coalescent Bayesian skyline prior. A sequence divergence rate of 0.024/Ma was used for *Cyt b* sequences (Yasuda et al. 2005). We assessed parameter convergence and visualized the skyline plots using Tracer v1.7.

Results

Morphological variation of *Micromys* species

A summary of the external and cranial measurements of *M. erythrotis*, *M. minutus*, and *Micromys* sp. is given in Table 1. *Micromys* sp. is much larger than *M. erythrotis* and *M. minutus* due to longer head-and-body and skull lengths. Most measurements of *M. minutus* highly overlap with those *M. erythrotis*. However, *M. minutus* has a longer UML compared to *M. erythrotis* (mean $\pm$ SD: 3.12 ± 0.06 mm, range span: 3.03–3.17 mm vs. mean $\pm$ SD: 2.90 ± 0.05 mm, range span: 2.86–2.99 mm).

The PCA based on 25 intact skulls produced three principal components with eigenvalues exceeding 1.0, which explained 75.437% of the variation (Table 2). The first principal component (PC1) explained 45.994% of the total variation and was strongly positively correlated with UTL, GLS, BL, NL, PIL, ML, and PPL (loading > 0.836), indicating that it represented the skull length. The second principal component (PC2) explained 19.161% of the total variation and was strongly positively correlated with CB and BMF (loading > 0.792). The third principal component (PC3) explained 10.282% of the total variation and was positively correlated with UML and M^2M^2 . A plot of PC1 and PC2 showed that *M. minutus* and *M. erythrotis* overlapped and occupied the negative field of both PC1 and PC2, and *Micromys* sp. samples mainly distributed in the positive region on PC1 (Fig. 1a), indicating that *Micromys* sp. had the longest skull. A plot of PC1 and PC3 showed that *M. erythrotis* are well separated from *M. minutus* along PC3 (Fig. 1b); *M. erythrotis* distributed in the negative corner regions of PC3, suggesting it had shorter upper molar row (UML) and narrower M^2M^2 .

Sequence information and phylogenetic analysis

We obtained 33 *Cyt b* [1140 bp], 32 *COI* [633 bp], 32 *RBP3* [1200 bp], 32 *GHR* [798 bp], and 31 *RAG 1* [1053 bp] sequences for the phylogenetic analyses. All new sequences have been deposited in GenBank (Accession Numbers: OQ561709–740, OQ576169–312; Table S1). For the three datasets, the topologies of ML and BI trees were highly similar, and only the BI gene trees are shown (Fig. 2). The phylogenetic analyses of all three gene datasets supported the monophyly of the genus *Micromys* (UFBoot ≥ 98 , PP = 1.00). The sister relationship between *Micromys* sp. and *M. erythrotis* was strongly supported in the BI tree of the concatenated nuclear gene trees (PP = 0.95) and concatenated mitochondrial-nuclear gene trees (PP = 0.99) and modestly in the ML tree of the two datasets (UFBoot = 93–94). However, this relationship was weakly supported in concatenated mitochondrial gene tree (UFBoot = 79, PP = 0.67).

Divergence time

Divergence time estimates showed that the most recent common ancestor (MRCA) of *Micromys* is from the Late Miocene ca. 11.01 Ma (95% CI = 9.75–12.29 Ma (Fig. 3). *Micromys minutus* diverged from *M. erythrotis* + *Micromys* sp. during the Pliocene ca. 3.35 Ma (95% CI = 2.12–5.04 Ma), after which *M. erythrotis* and *Micromys* sp. diverged from each other in the late Pliocene ca. 2.55 Ma (95% CI = 1.45–3.75 Ma).

Species delimitation and species tree

The genetic distances between the three *Micromys* species based on *Cyt b* ranged from 11.4% to 14.1%, while *COI*-based distances ranged from 9.2% to 11% (Table 3). Under different algorithms, models, and priors, BPP consistently supported all three *Micromys* taxa as distinct (PP = 1.0) (Table S6). The GMYC analysis based on mitochondrial-nuclear genes also recognized three putative species (95% CI = 3–9), with a basal split between *M. minutus*, *M. erythrotis*, and *Micromys* sp.

Geographical distribution structure of genus *Micromys*

The phylogenetic tree based on 140 *Cyt b* sequences strongly supported three main branches in *Micromys*, representing *M. minutus*, *M. erythrotis*, and *Micromys* sp. (Fig. 4a). The topology of the *Cyt b* haplotype network was similar to the phylogenetic tree (Fig. 4b). A total of 46

Table 1 Summary statistics (mean, standard deviation, ranges) of external and skull measurements (in millimeters) of *Micromys* specimens used in the study

Skull measures	<i>M. minutus</i> (N=3)	<i>M. erythrotis</i> (N=4)	<i>Micromys</i> sp. (N=20)
W	6.70±0.36 6.2–7.0 (3)	6.14±1.02 4.8–7.5 (4)	8.40±2.34 3.0–14 (20)
HB	54.0±0.82 53–55 (3)	57±5.10 51–65 (4)	65.2±5.02 54–79 (20)
TL	66.3±2.87 63–70 (3)	68.8±7.95 61–82 (4)	69.0±6.91 57.0–80.0 (20)
HL	14.8±0.24 14.5–15 (3)	14.25±0.83 13.0–15.0 (4)	14.7±0.81 13.0–16.0 (20)
EL	10.0±0 10.0–10.0 (3)	9.88±0.74 9.0–11.0 (4)	10.8±1.08 9.0–13.0 (20)
GLS	17.61±0.46 17.28–18.26 (3)	17.94±0.59 17.22–18.83 (4)	19.05±0.57 18.08–20.1 (20)
BL	16.32±0.23 16.14–16.65 (3)	16.02±0.35 15.55–16.47 (4)	17.32±0.53 16.43–18.39 (20)
PIL	8.76±0.23 8.54–9.08 (3)	8.72±0.27 8.41–9.14 (4)	9.42±0.27 8.85–10.07 (20)
PPL	5.70±0.08 5.61–5.81 (3)	5.87±0.41 5.29–6.35 (4)	6.21±0.56 5.46–7.26 (20)
ZB	9.21±0.25 8.94–9.54 (3)	9.21±0.21 8.92–9.51 (4)	9.58±0.33 8.96–10.27 (20)
CB	9.24±0.10 9.16–9.39 (3)	9.07±0.20 8.85–9.38 (4)	9.37±0.26 8.71–9.92 (20)
IOB	3.34±0.02 3.32–3.36 (3)	3.34±0.13 3.2–3.52 (4)	3.22±0.10 3.04–3.37 (20)
M ² -M ²	3.68±0.02 3.65–3.71 (3)	3.62±0.08 3.51–3.72 (4)	3.80±0.08 3.64–3.96 (20)
RB	2.96±0.07 2.86–3.02 (3)	3.06±0.08 2.95–3.18 (4)	3.25±0.12 3.06–3.55 (20)
UTL	7.99±0.07 7.91–8.09 (3)	8.07±0.22 7.81–8.38 (4)	8.52±0.23 8.14–9.11 (20)
UML	3.12±0.06 3.03–3.17 (3)	2.90±0.05 2.86–2.99 (4)	3.01±0.09 2.83–3.22 (20)
NL	5.29±0.21 4.99–5.47 (3)	5.44±0.06 5.37–5.52 (4)	5.89±0.29 5.50–6.68 (20)
CH	7.09±0.15 6.93–7.28 (3)	7.23±0.20 7.01–7.48 (4)	7.51±0.26 7.2–8.23 (18)
BMF	3.62±0.09 3.54–3.75 (3)	3.56±0.16 3.35–3.77 (4)	3.66±0.14 3.35–3.91 (20)
ML	8.45±0.07 8.35–8.53 (3)	8.62±0.24 8.36–8.97 (4)	9.30±0.30 8.68–9.97 (20)
LTR	6.68±0.03 6.65–6.72 (3)	6.47±0.12 6.26–6.56 (4)	6.89±0.19 6.63–7.52 (20)

The sample sizes (N) are also shown

Refer to the "Material and methods" section for the meaning of the feature abbreviations

haplotypes were identified, including 26 for *M. minutus*, 13 for *Micromys* sp., and 7 for *M. erythrotis*.

Micromys minutus had the most haplotypes, clustered into five major clades: Europe, Japan-Korea, Russia, Anhui and Taiwan Island. The Europe group comprised eight

haplotypes arranged in a star-like network, with H15 (samples from Germany, Portugal, Finland, and Denmark) as the central haplotype. The Japan-Korea clade comprised nine haplotypes and was also arranged in a star-like network, with H6 (samples from Japan) or H8 (samples from Nagasaki,

Table 2 Craniodental variation in *Micromys* based on principal component analysis (PCA)

Variables	Principal component		
	1	2	3
UTL	0.958	0.026	0.097
GLS	0.951	0.130	-0.040
BL	0.858	0.175	0.213
NL	0.857	0.100	-0.016
PIL	0.839	0.206	0.080
ML	0.838	0.329	0.093
PPL	0.836	-0.200	-0.320
ZB	0.638	0.590	-0.129
LTR	0.634	0.367	0.342
IOB	-0.583	0.390	-0.171
CB	0.121	0.860	0.063
BMF	-0.190	0.792	0.029
RB	0.606	0.652	-0.048
CH	0.478	0.532	0.015
UML	-0.178	-0.201	0.899
M ² M ²	0.368	0.257	0.702
Eigenvalue	7.359	3.066	1.645
Variance explained	45.994	19.161	10.282

The table shows factor loadings, eigenvalues, and percentage of variance explained by the first three components (PC1, PC2, and PC3) from a PCA of 16 log₁₀-transformed measurements

The meanings of these abbreviations are given in the "Material and methods" section

Japan, and South Korea) as central haplotypes. The MJ network showed a close relationship between the Europe and Japan-Korea groups. The Russia clade was split into the Far East region and Novosibirsk, with H17 being the central haplotype of the Novosibirsk population. The other two clades from Anhui and Taiwan Island were most closely related to the Novosibirsk population. The MJ network showed that samples from Russia and China, may have originated from

the same unobserved intermediate haplotype (haplotype A, Fig. 4a), closely related to another unobserved intermediate haplotype (haplotype B, Fig. 4a) comprised of samples from between Japan-Korean and European populations.

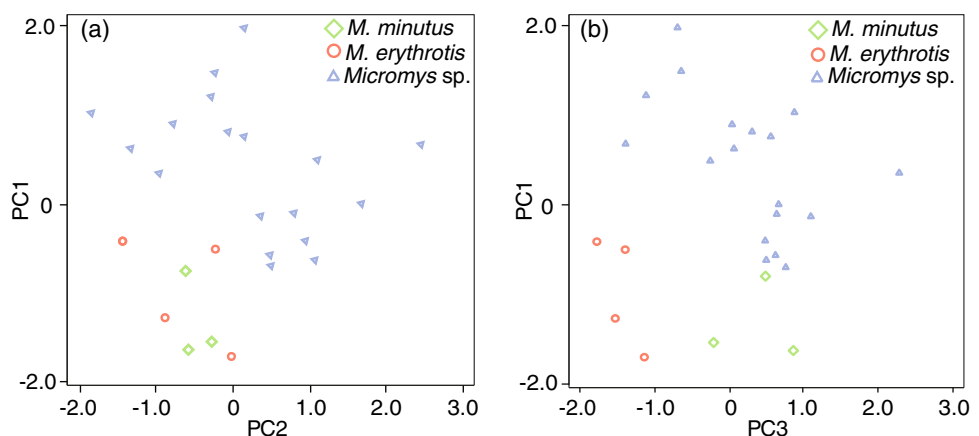
The haplotypes of *M. erythrotis* were divided into two clades. The first clade includes samples from Vietnam, Yunnan, Guizhou, Anhui, and Hunan, China. Another clade includes samples from Shaanxi and Sichuan, China. The haplotypes of *Micromys* sp. were divided into three clades. The first clade included samples from Yunnan, China (Dali, Kunming, and Lushui), the second clade included samples from Yunnan, Guizhou, China (Kunming and Zhao-tong), and the third clade included samples from Yunnan (Yongsheng), Sichuan (Kangding), and Tibet, China.

Distribution mapping showed that *M. minutus* is the most widely distributed species, with populations spanning Europe, Russia, Japan, Korea, and eastern China (Fig. 5). *Micromys erythrotis* is distributed in North Vietnam, and central-southern China, including Yunnan, Sichuan, Shaanxi, Hunan, and Anhui. The species is sympatric with *M. minutus* in Qingliang Mountain, Anhui. *Micromys* sp. is distributed in western Sichuan, north-west Yunnan, western Guizhou, and Southeast Tibet, where it is restricted to the middle-high elevations of the Himalayas and Hengduan Mountains (1600–3800 m). It is sympatric with *M. erythrotis* in Bijie, Guizhou, China.

Demographic history and range shifts

The Fu's FS and Tajima's D tests of *M. minutus* were negative; the Fu's FS value was statistically significant ($P < 0.05$) (Table S7). Although the mismatch distribution test of *M. minutus* was multimodal, the SSD and r values were not significant (Fig. 6; Table S7), indicating that it may have experienced population expansion. Overall, Fu's FS, Tajima's D, and mismatch distribution tests showed no explicit signals for population changes in *M. erythrotis* and *Micromys* sp.

Fig. 1 Craniodental variation in *Micromys* based on principal component analysis (PCA). The scatter plots show the morphospace among *M. minutus*, *M. erythrotis*, and *Micromys* sp. based on the first three PCA components (PC1, PC2, and PC3) from an analysis of 16 log₁₀-transformed craniodental measurements. PC1 vs. PC2 is shown in (a), and PC1 vs. PC3 in (b)



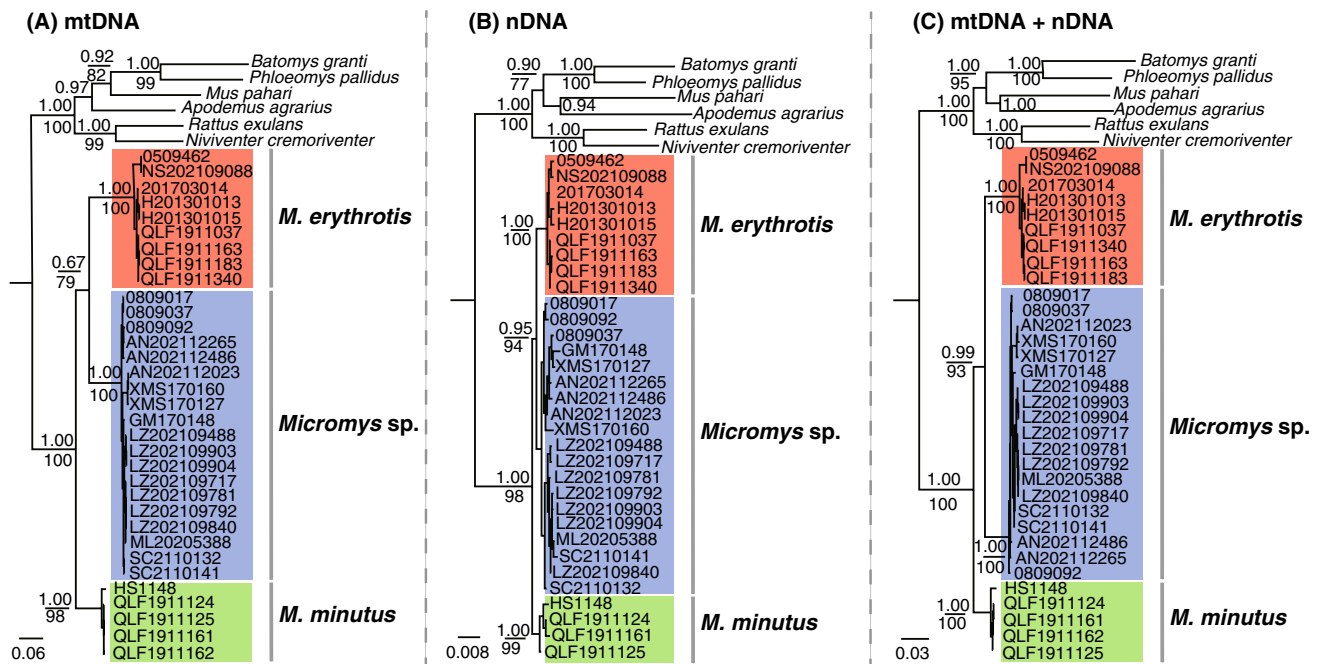


Fig. 2 The phylogenetic associations between *Micromys* species inferred using Bayesian inference (BI). The figure shows phylogenetic trees derived from **A** the concatenated mitochondrial genes, **B** the

concatenated nuclear genes, and **C** the concatenated mitochondrial and nuclear genes. The branch numbers indicate BI posterior probabilities (above) and ultrafast bootstrap values (below)

Fig. 3 The divergence time of *Micromys* species inferred from a time-calibrated phylogeny based on two mitochondrial and three nuclear genes in BEAST v2.6.7. The branch lengths represent time, node bars indicate the 95% CI for each clade age, and red bars indicate the calibration points. The node numbers represent posterior probabilities (upper) and the median ages of the divergence times (below)

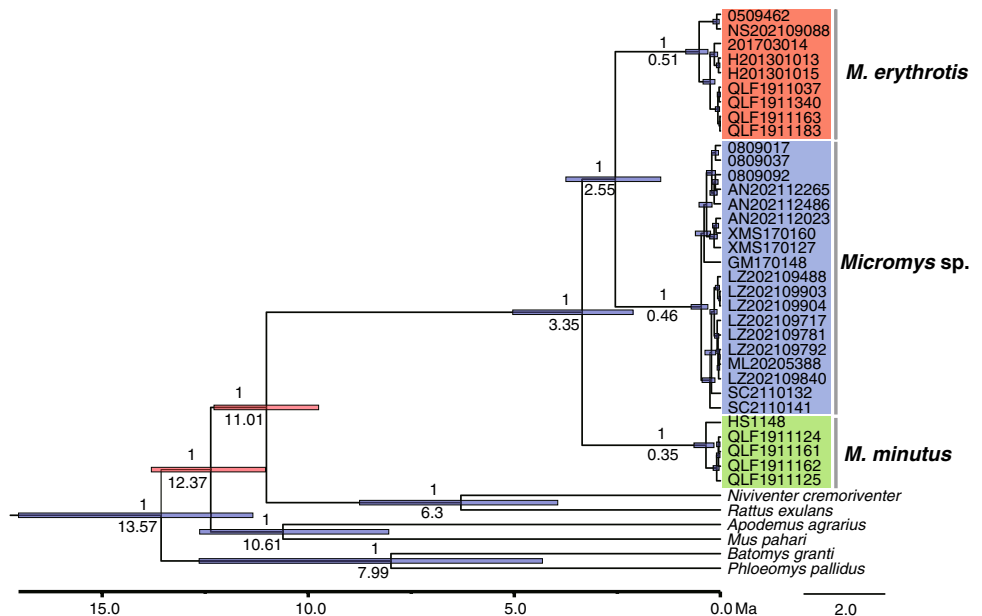
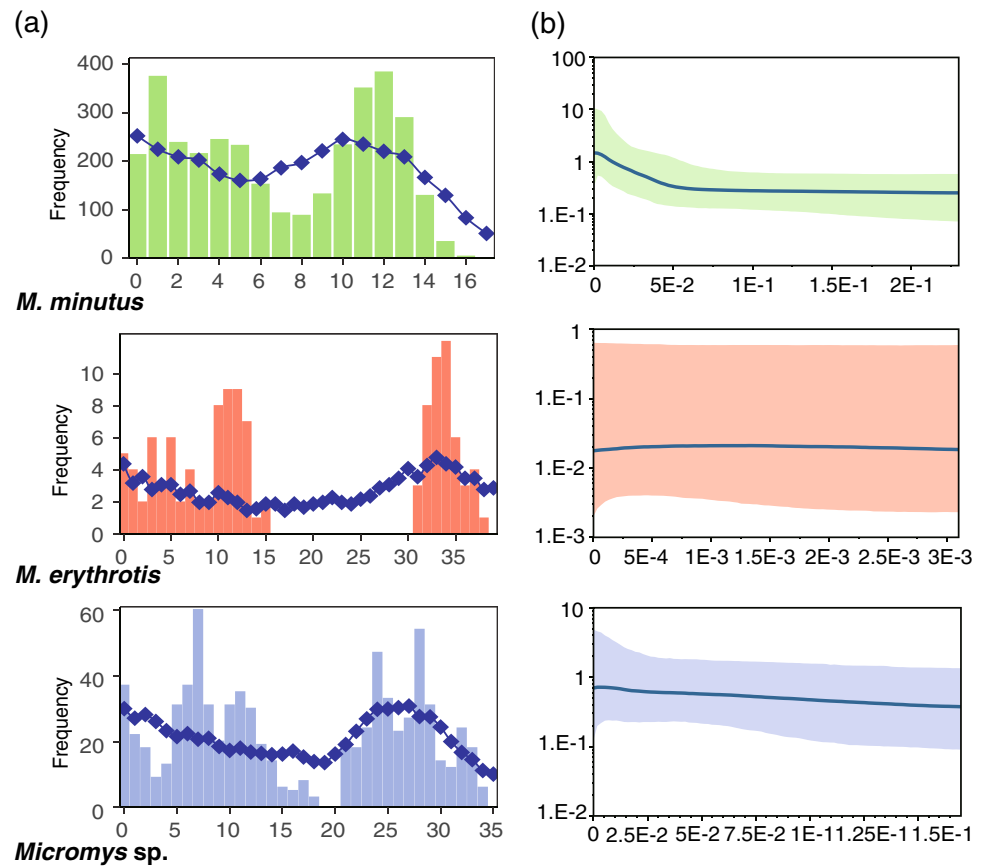


Table 3 Genetic distance between *Micromys* species based on *Cyt b* (in the lower triangle) and *COI* (in the upper triangle) genes

<i>Cyt b</i> / <i>COI</i>	<i>M. minutus</i>	<i>M. erythrotis</i>	<i>Micromys</i> sp.
<i>M. minutus</i>		0.110	0.095
<i>M. erythrotis</i>	0.141		0.092
<i>Micromys</i> sp.	0.139	0.114	

Bayesian skyline plots (BSP) revealed a relatively explicit demographic history for the three species in the genus *Micromys* (Fig. 6). *Micromys minutus* showed significant population size increases since 50 kyr BP. *Micromys erythrotis* was nearly at a stable population size, and *Micromys* sp. showed a slight trend of increasing population size since 170 kyr BP.

Fig. 6 Historical demography of three *Micromys* species inferred from *Cyt b* sequences. **a** Mismatch distribution analysis based on pairwise nucleotide sequences. The bars represent the observed value and the squares represent the simulated value. **b** Bayesian skyline plots showing effective population size as a function of time. The upper and lower limits of light colors trend represent the 95% confidence intervals of HPD analysis



Discussion

Being one of the most widespread and smallest-sized rodents likely contributed to the controversial taxonomy and distribution of *Micromys*, with most studies recognizing *M. minutus* as the only species of the genus (Wilson and Reeder 2005; Wilson et al. 2017). Recently, several studies suggested *M. erythrotis* should be a second valid *Micromys* species (Yasuda et al. 2005; Abramov et al. 2009; Pei et al. 2021). In this study, the results of phylogenetic trees indicated that *Micromys* was divided into three clades, representing *M. minutus*, *M. erythrotis*, and *Micromys* sp. The genetic distances between the three clades range from 11.4% to 14.1% based on *Cyt b* and 9.2% to 11% based on *COI* (Table 3). The values are higher than the distances between many sister species in other rodent genus, such as *Apodemus* (e.g., *A. agrarius* and *A. chevrieri*, *Cyt b*: 7.4%, *COI*: 6.9; Ge et al. 2019) and *Niviventer* (e.g., *N. huang* and *N. mekongis*, *Cyt b*: 8.0%, *COI*: 5.2%; Ge et al. 2021), and are higher than the average genetic distance between sister species of mammals (*Cyt b*: 8.1%; Bradley & Baker 2001). The large genetic distance and morphological differences between the three clades supported the species status of *M. erythrotis* and *Micromys* sp. The *Micromys* sp. samples were collected at high altitudes in Yunnan, western Sichuan, western

Guizhou, and Southeast Tibet. Several *Micromys* species and subspecies have been described in this region, including *Mus* (= *Micromys*) *pygmaeus* Milne-Edwards 1874 (type locality: Moupin, Sichuan, China), *M. m. berezowskii* Argyropulo, 1929 (type locality: Pingwu, northern Sichuan, China), and *M. m. pianmaensis* Peng, 1981 (type locality: Pianma, Yunnan, China). The morphological characteristics of *Micromys* sp. are consistent with the descriptions of *M. pygmaeus* in Milne-Edwards (1874). And one of the *Micromys* sp. collection sites (Mugecuo, Kangding, Sichuan) is close to the type locality of *M. pygmaeus*. Thus, we believe that *Micromys* sp. belongs to *Mus pygmaeus*, the oldest name of Chinese *Micromys*, and propose *M. pygmaeus* (Milne-Edwards 1874) as the third species of the genus *Micromys*. Our study included the specimens collected from the type locality of *M. m. pianmaensis*, Pianma, Yunnan, China, and revealed that it is the same species as *M. pygmaeus*. Although we did not examine *berezowskii* specimens, the available *COI* sequences (Li et al. 2015) showed that specimens from Beichuan, northern Sichuan (near Pingwu, type locality of *M. m. berezowskii*) belong to *M. erythrotis* (results are not shown), implying that *M. m. berezowskii* is likely a synonym of *M. erythrotis*.

Our divergence estimates indicated that *Micromys* originated during the Late Miocene (Fig. 3), consistent with

earlier studies (Pagès et al. 2016; Aghová et al. 2018). The fossil records of at least ten *Micromys* species were already diverging through Eurasia by the end of the Miocene, including in Inner Mongolia, Gansu, and Hebei in China, Greece, Spain, and Russia by the end of the Miocene (Table S5). These fossil species do not represent a single lineage leading to the extant *M. minutus* (Storch and Dahlmann 1995), suggesting that the genus underwent radiations during the Miocene and a massive species extinction event in the Pliocene (Yasuda et al. 2005). The uplift of the Tibetan Plateau and Hengduan Mountains and the global cooling and desiccating events at the Pliocene/Pleistocene boundaries appear to have played critical roles in the split of *Micromys*. The split between *M. minutus* and *M. erythrotis* + *M. pygmaeus* occurred in the Pliocene ca. 3.35 Ma (95% CI = 2.12–5.04 Ma). During the period ca. 3.4 Ma, a large-scale rapid uplift occurred in the Tibetan Plateau and the Hengduan area, breaking up the level of the Hengduan Mountains (Chen 1992, 1996; Li and Fang 1999). The rapid uplift of the Hengduan mountains and ensuing climate change may be one of the most significant drivers of the split of *M. minutus* and *M. erythrotis* + *M. pygmaeus*. *Micromys erythrotis* and *M. pygmaeus* diverged in the early Pleistocene at about 2.55 Ma (95% CI = 1.45–3.75 Ma). During this period, global cooling and desiccating events, which are well-documented (Webb and Bartlein 1992; Fujiki and Ozawa 2008; Lunt et al. 2008), and the uplift of the Qinghai–Tibet Plateau (Li and Fang 1999; Shi et al. 1999) have together changed the habitats of the surrounding area. These global climatic changes around the Pliocene/Pleistocene boundary and the uplift of the Qinghai–Tibet Plateau probably drove the split of *M. erythrotis* and *M. pygmaeus*.

The Fu's FS, Tajima's D, mismatch tests, and the Bayesian skyline plots revealed that *M. minutus* underwent a recent population size increase (Fig. 6; Table S7). The oldest fossils of *M. minutus* were found almost simultaneously in Russia and Spain in the Pliocene. Most of the fossils in China were found in the Pleistocene, while no fossils have been found in Japan (Table S5). Therefore, the *M. minutus* of the Japan–Korea region and eastern China are more likely migrants. Consistent with Yasuda et al. (2005), our haplotype network showed a close relationship between *M. minutus* populations from Europe and those from the Japan–Korea region and between populations from Russia and eastern China (Fig. 4). Yasuda et al. (2005) assumed that European and East Asian populations were derived from a refugium, perhaps via a corridor to the north of a simultaneously expanding central Siberian population. Our results suggest a north–south migration route via eastern China, which allowed *M. minutus* to reach Taiwan Island.

It is worth noting that some *M. minutus* fossils from southwestern China may have been misidentified due to taxonomic disagreements. For example, the “*M. minutus*”

fossil from Chenggong, Yunnan (Qiu et al. 1984) and *M. liui* fossil from Guizhou, China (Zheng 1993) have a stout mandible with a large ramus region and condyloid process, which appeared more closely related to *M. pygmaeus* than *M. minutus*. Thus, the historical distribution of *M. minutus* may also have been restricted to eastern China, while the southwestern parts of China were occupied by *M. erythrotis* and *M. pygmaeus*.

Based on morphological and molecular analyses, our results strongly support that the genus *Micromys* contain three distinct species: *M. minutus*, *M. erythrotis*, and *M. pygmaeus*. We provide a short review of each species below.

Taxonomic accounts

Micromys minutus (Pallas 1771)

Common name: Eurasian Harvest Mouse; 巢鼠.

Mus minutus Pallas 1771: 453–504. Type locality: Ulyanovsk Oblast, Russia.

Mus soricinus Hermann, 1780: 61. Type locality: Strasbourg, Bas-Rhin, Eastern France.

Mus triticeus Boddaert, 1785: 111. Type locality: Hampshire, England.

Mus minimus White, 1789: 43. Type locality: Selborne, Hampshire, England.

Mus messorius Kerr, 1792: 230. Type locality: Hampshire, England.

Mus minutus flavus Kerr, 1792: 232. Type locality: Siberia.

Mus avenarius Wolf, 1794: 315. Type locality: Europe.

Mus pendulinus Hermann, 1804: 61. Type locality: Strasbourg, France.

Mus parvulus Hermann, 1804: 62. Type locality: Strasbourg, France.

Mus arvensis Leach, 1816: 7. Type locality: Devonshire, England.

Mus campestris Desmarest, 1822: 543. Type locality: France.

Mus pratensis Ockskay, 1831: 243. Type locality: Western Hungary.

Mus minatus Schinz, 1840. Type locality: Europe.

Micromys agilis Dehne, 1841: 16. Type locality: Dresden, Germany.

Mus oryzivorus de Sclys-Longchamps, 1841: 247. Type locality: Lombardy, Italy.

Mus pumilus F. Cuvier, 1842. Type locality: Paris, France.

Mus meridionalis Costa, 1844: 33. Type locality: Naples, Italy.

Mus arundinaceus Petenyi, 1882: 91. Type locality: Budapest and Western Hungary.

Mus minutus ussuricus Barrett-Hamilton, 1899: 344. Type locality: Ussuri region, South-eastern Siberia.

Micromys minutus japonicus Thomas, 1906: 351. Type locality: Kyushu, Japan.

Mus minutus batawvi Kastschenko, 1910: 284. Type locality: Transbaikalia, Eastern Siberia.

Mus minutus kyzmanovi Kastschenko, 1910: 284. Type locality: Transbaikalia, Eastern Siberia.

Mus minutus fenniae Hilzheimer, 1911: 15. Type locality: Mantsala, Finland.

Mus minutus sareptae Hilzheimer, 1911: 18. Type locality: Sarepta, Lower Volga, Russia.

Micromys minutus kastschenkoi Charlamagne, 1915: 69. Type locality: Ukraine.

Micromys minutus aokii Kuroda 1922: 42–45. Type locality: Tsushima, Japan.

Micromys minutus mehelyi Bolkay, 1925: 12. Type locality: Bosnia, Yugoslavia.

Micromys minutus brauneri Martino, 1930: 60. Type locality: Kraljevo, Serbia, Yugoslavia.

Micromys minutus hondonis Kuroda 1933: 243. Type locality: Hondo, Japan.

Micromys minutus subobscurus Fritzsche, 1934: 431. Type locality: Westmond, Germany.

Micromys minutus takasagoensis Tokuda, 1941: 78. Type locality: Sikayau, Formosa (= Taiwan).

Micromys minutus kurodai Mori, 1942: 77. Type locality: Jilin, China.

Micromys minutus hertigi Johnson and Jones, 1955: 167. Type locality: Cheju Do, Korea.

Remarks: *Micromys minutus* (Pallas 1771) is a small-sized species in the genus (mean $\pm$ SD of HB: 54.0 ± 0.82 mm; mean $\pm$ SD of GLS: 17.61 ± 0.46 mm; Table 1). The species tail is shorter than the head and body length in European specimens but longer than the head and body length for specimens from East Asia (Japan and China) (Kuroda 1922, 1933; Cross 1967; Corbet and Hill 1992; Table 1). Its

dorsal and ventral pelages are bicolored; the dorsal pelage is yellowish with orange-brown tones and contrasts the pure ventral underparts, a coloration that distinguishes it from *M. erythrotis* and *M. pygmaeus*. It has pale yellow hairs on the dorsal surfaces of hands and hind feet and a bicolored tail that is brown on the upper side and grizzled on the lower side (Fig. 7). Its mandible is weaker, and the condyloid process is smaller than that of *M. erythrotis* and *M. pygmaeus* (Fig. 8). Despite *M. minutus*'s wide distribution, there is little intraspecific genetic and karyotypic variation across regions (Orlov and Bulatova 1983; Yasuda et al. 2005).

Distribution: It is widely distributed in Europe and North-East Asia (Fig. 5). In Europe, it has been recorded in most countries, including France, Finland, Norway, Austria, Germany, Italy, Sweden, and Russia. In Asia, the species has been recorded in Japan, Korea, and northern and eastern China. Most specimens were recorded in lowlands at elevations below 1700 m. It is sympatric with *M. erythrotis* in Jixi, Anhui, China, at an elevation of 1600 m.

Micromys erythrotis (Blyth 1856)

Common name: Red-eared Harvest Mouse; 红耳巢鼠.

Mus erythrotis Blyth 1856: 721. Type locality: Assam, India.

Micromys minutus berezowskii Argyropulo, 1929: 253. Type locality: Pingwu, Sichuan, China.

Micromys minutus shenshiensis Li, Wu and Shao, 1965. Type locality: Shannxi.

Micromys minutus zhenjiangensis Huang, 1989: 10–13. Type locality: Zhenjiang, Jiangsu, China.

Remarks: *Micromys erythrotis* is similar to *M. minutus* in size, but much smaller than *M. pygmaeus* (Table 1). It has long, dense fur with no clear transition in dorsal and ventral color. Its dorsal pelage is dark russet with a rich dark russet

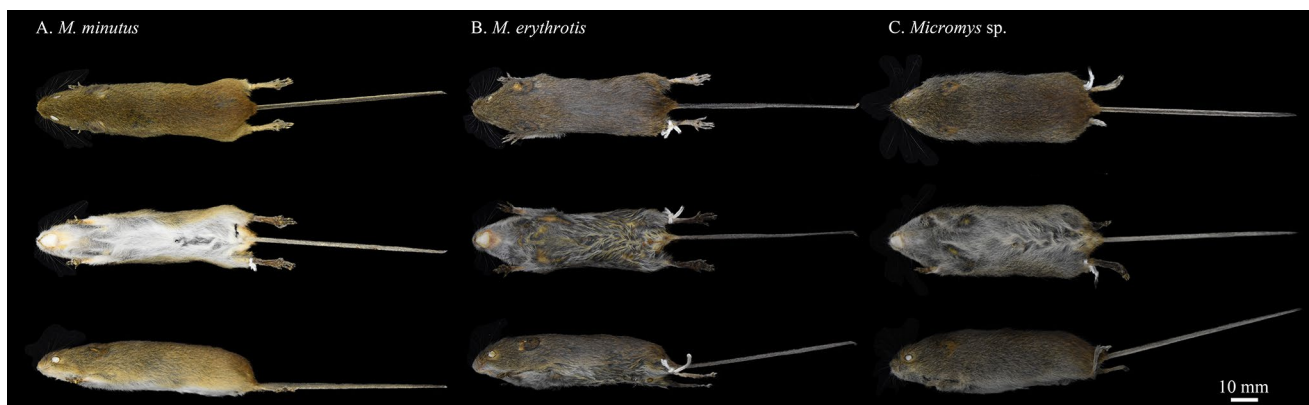


Fig. 7 The ventral, dorsal, and lateral views of the skin of **A** *M. minutus* (QLF1911125), **B** *M. erythrotis* (QLF1911183) and **C** *M. pygmaeus* (LZ202109960)

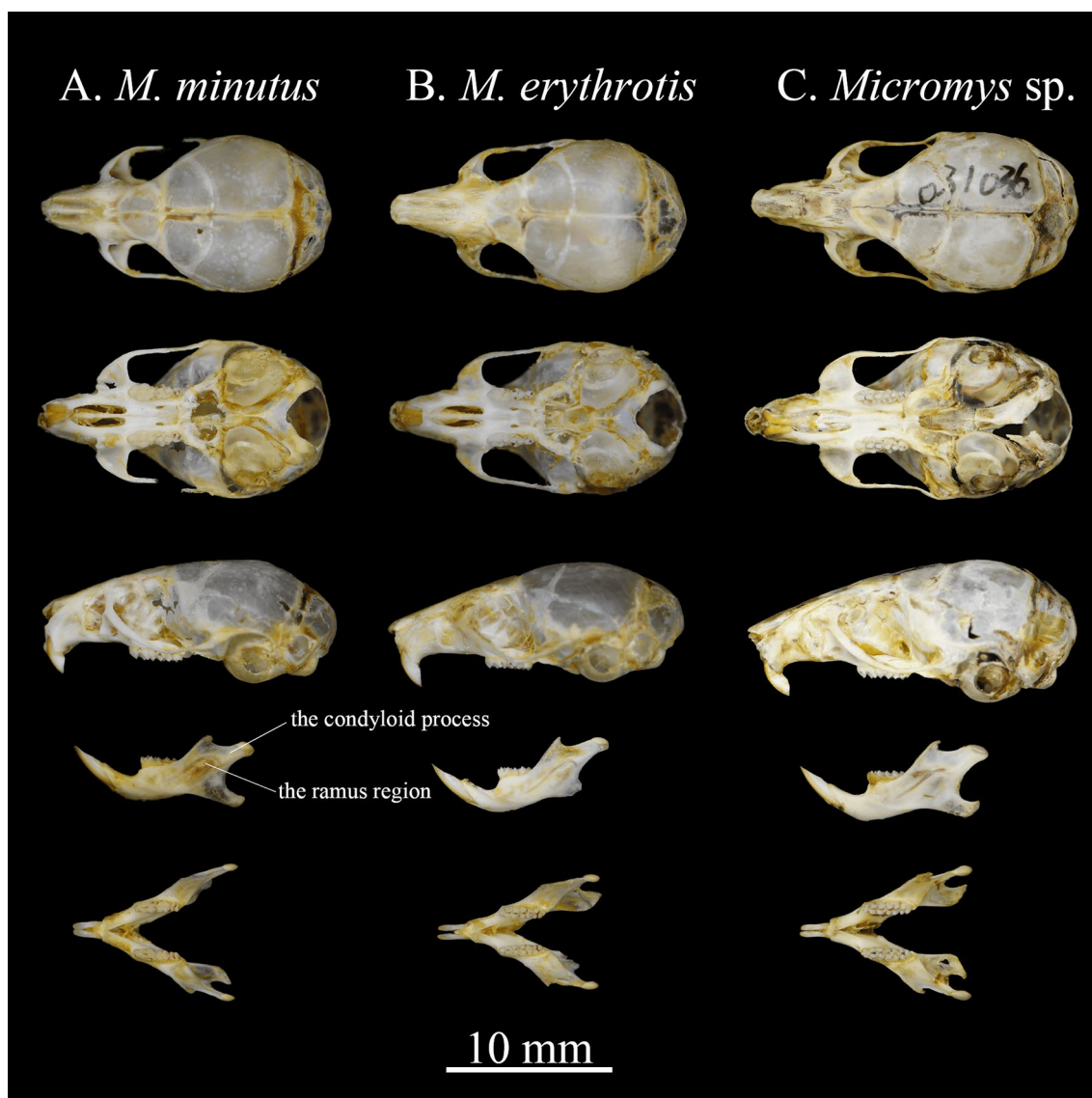


Fig. 8 The ventral, dorsal, and lateral views of the skull and mandibles of **A** *M. minutus* (QLF1911162), **B** *M. erythrotis* (QLF1911183), **C** *M. pygmaeus* (0809092)

and evenly lined with black color and bright rufous short hairs on the ears and around the rump (Fig. 7), while the ventral pelage is slate grey. Its tail length is longer than the head and body, hirsute, and is not clearly bicolored. It has a weak mandible, a long and thin condyloid process, and a lower toothrow shorter length than in *M. minutus* and *M. pygmaeus* (Fig. 8).

Micromys erythrotis was originally described based on a single specimen from Assam, India (Blyth 1856). The tail length of the type specimen was slightly longer than the head and body (body length 57.2 mm, tail length 60.3 mm). Corbet and Hill (1992) reported a specimen with a very long tail length from Assam (body length 60 mm, tail length 90 mm). It previously included *M. pygmaeus* and *M.*

minutus berezowskii as synonyms (Ellerman and Morrison-Scott 1951). Our results support that *M. pygmaeus* is a distinct species. The available *COI* sequences showed that the specimens from Beichuan, northern Sichuan (near the type locality of *M. m. berezowskii*) belonged to *M. erythrotis*. Therefore, we tentatively assigned *M. m. berezowskii* to *M. erythrotis* rather than to *M. pygmaeus* like Allen (1940) did. Currently, there is little information on the subspecies *M. m. zhenjiangensis* from Zhenjiang, Jiangsu, China; thus, we retained it in *M. erythrotis* following Wei (2022).

Distribution: The species is known from Vietnam, Myanmar, India, and central and southern China, including Hunan, Hubei, Shannxi, Anhui, Jiangsu, Shanghai, Fujian,

Guangdong, Yunnan, Sichuan, and Guizhou. Most of our specimens were collected from mountain areas below 1700 m.

Micromys pygmaeus (Milne-Edwards 1874)

Suggested common name: Pygmy Harvest Mouse; 高原巢鼠.

Mus pygmaeus Milne-Edwards 1874: 291. Type locality: Moupin (Baoxing), Sichuan, China.

Micromys minutus pianmaensis Peng, 1981:167–176. Type locality: Pianma, Yunnan, China.

Remarks: *Micromys pygmaeus* is the biggest species in the genus *Micromys* (mean $\pm$ SD of HB: 65.2 ± 5.02 mm; mean $\pm$ SD of GLS: 19.05 ± 0.57 mm; mean $\pm$ SD of CH: 7.51 ± 0.26 mm; Table 1). It has a pelage similar to *M. erythrotis*, but with longer and softer hairs dark russet on the dorsal surface (Fig. 7). The dorsal surface is lined with some even and minute black hairs, and the ears and around the rump have bright rufous short hairs. Its ventral pelage is slate gray, and the tips of the hairs are dull white, while the dorsal surfaces of hands and hind feet are covered with white hairs that are much longer than in *M. erythrotis*. Its tail is bicolored and covered with short hairs, blackish above and paler beneath. Its mandible is deep and stout, with the ramus region and condyloid process are much larger than in *M. minutus* and *M. erythrotis* (Fig. 8).

Micromys pygmaeus was first described by Milne-Edwards (1874) under the genus *Mus*. Consequent works often regarded it as a subspecies or synonym of *M. minutus* or *M. erythrotis* (Allen 1940; Ellerman and Morrison-Scott 1951; Wilson and Reeder 2005; Wei 2022). While *M. m. berezowskii* was regarded as a synonym of *M. pygmaeus* by Allen (1940), it is more likely a synonym of *M. erythrotis* (see the remarks on *M. erythrotis*). Our analyses also indicated that *M. m. pianmaensis* should be a synonym of *M. pygmaeus*. The species is restricted to the middle-to-high elevations of the Himalayas and Hengduan Mountains, where its longer hairs relative to other *Micromys* species may be an adaptation to the cold climate of the high elevations.

Distribution: It is currently only known from middle and high elevations (1600–3800 m) in southwestern China, including western Guizhou, western Sichuan, north-west Yunnan, and southeast Tibet (Fig. 5). It is sympatric with *M. erythrotis* in Bijie, Guizhou, China.

External morphological key to *Micromys* species

1. a. The dorsal pelage is yellowish with orange-brown tones and the ventral pelage is pure white *M. minutus*.

b. The dorsal pelage is dark russet lacking the orange-brown tones and the ventral pelage is slate grey 2.

2. a. Small; LTR < 6.6 mm; mandible is weak with a long and thin condyloid process *M. erythrotis*.

b. Large; LTR > 6.6 mm; mandible is stout with a robust condyloid process *M. pygmaeus*.

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Author contributions ZYC and XLJ conceived the ideas and designed methodology; ZYC, XXP, SJL, ZFS and QL collected the data; ZYC and XXP analysed the data; ZYC and XXP led the writing of the manuscript. KOO contributed to editing the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

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Data availability The datasets generated during the present study are available from the author on request.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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